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Department of Computer Science
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Computer Vision - AI3302
2nd semester 2026
Project

Report

Discovering Emergent Visual Identities in Static Images

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1. Introduction

Interior design images contain rich visual information related to color, texture, layout, lighting, and overall composition. These visual elements often reflect meaningful design patterns, but they are not always fully captured by predefined style labels. In many cases, different styles may overlap visually, while images belonging to the same label may still vary in appearance. This makes it difficult to rely only on traditional categories when analyzing such data.

Traditional computer vision approaches often focus on supervised classification or predefined labels, which limits their ability to discover hidden visual structures in complex image collections. This creates a gap in current systems, especially when dealing with visually rich domains such as interior design, where broader aesthetic relationships may exist beyond fixed style categories.

To address this challenge, this project explores the use of unsupervised learning techniques in computer vision to discover hidden visual identities in static interior design images. Instead of predicting existing labels, the system groups images based on visual similarity, allowing it to uncover patterns that may not be explicitly defined in the dataset. The proposed solution analyzes images using a hybrid visual representation that combines deep features extracted from ResNet50 with handcrafted color histogram and texture descriptors. These features are then reduced using Principal Component Analysis (PCA) and grouped through clustering methods to reveal meaningful visual patterns.

The main objective of this project is to discover meaningful emergent visual identities from interior design images and evaluate how well these identities reflect actual visual patterns in the data. By combining quantitative metrics with qualitative interpretation, the project aims to provide a more practical and interpretable understanding of clustering in visually complex datasets.

2. Related Work

Recent advances in computer vision have significantly improved image analysis through the use of deep learning techniques. Early work by Alex Krizhevsky et al. [1] demonstrated the effectiveness of deep convolutional neural networks (CNNs) for large-scale image classification tasks, establishing a strong foundation for visual feature extraction.

Building on this, Kaiming He et al. [2] introduced residual networks (ResNet), which enable the training of very deep neural networks by addressing the vanishing gradient problem. This advancement allows for more powerful and discriminative feature representations, which are widely used in modern computer vision systems.

In the context of unsupervised learning, recent work such as DeepCluster (Caron et al.) [3] explores the integration of clustering with deep feature learning. This approach iteratively groups visual features using clustering algorithms such as K-Means and uses these groupings to improve feature representation without relying on labeled data.

Furthermore, visualization techniques such as t-SNE, proposed by Laurens van der Maaten and Geoffrey Hinton [4], have been widely used to analyze high-dimensional data. By projecting features into a lower-dimensional space, t-SNE helps reveal hidden structures and relationships between image clusters.

Dimensionality reduction methods have also played an important role in machine learning pipelines. Jolliffe and Cadima [5] reviewed Principal Component Analysis (PCA) as an effective technique for reducing high-dimensional data while preserving most of the important variance. PCA improves interpretability, removes redundancy, and reduces computational complexity, making it highly suitable for preprocessing before clustering tasks.

Traditional handcrafted feature extraction methods remain valuable in many image analysis applications. Rahim et al. [6] proposed the use of Local Binary Patterns (LBP) for recognition systems, demonstrating that LBP histograms can efficiently capture texture information and represent visual similarity between images.

In clustering research, Kodinariya and Makwana [7] reviewed several methods for determining the optimal number of clusters in K-Means clustering, including the Elbow Method, Silhouette Score, and other statistical approaches. Their study highlighted the importance of selecting an appropriate number of clusters to improve clustering quality and interpretability.

In the domain of interior design, Jinsung Kim and Jin-Kook Lee [8] proposed a deep learning-based approach for recognizing interior design styles using reference images. Their method utilizes a pre-trained VGG16 model to classify design styles, demonstrating the effectiveness of data-driven approaches in style recognition tasks.

In addition to deep learning features, traditional handcrafted feature extraction techniques have also been widely used in image analysis tasks. Color histograms are commonly utilized to represent color distribution and visual appearance within images, while Local Binary Pattern (LBP) descriptors are effective for capturing texture information and surface patterns. These techniques provide complementary low-level visual characteristics that may not always be fully captured by deep neural networks alone.

Unlike previous approaches, this project adopts an unsupervised learning strategy to discover visual identities directly from images. The proposed method combines deep semantic representations extracted using ResNet50 with handcrafted color histogram and LBP texture descriptors. PCA is then applied to reduce feature dimensionality before clustering. Finally, K-Means is used to group visually similar images into emergent identities. This hybrid feature representation enables the system to capture both high-level semantic information and low-level visual patterns.

3. Proposed Methodology & Implementation

3.1 System Pipeline Diagram

To provide a clear overview of the proposed approach, the figure below illustrates the main stages of the system pipeline, from data preparation to clustering, evaluation, and inference.

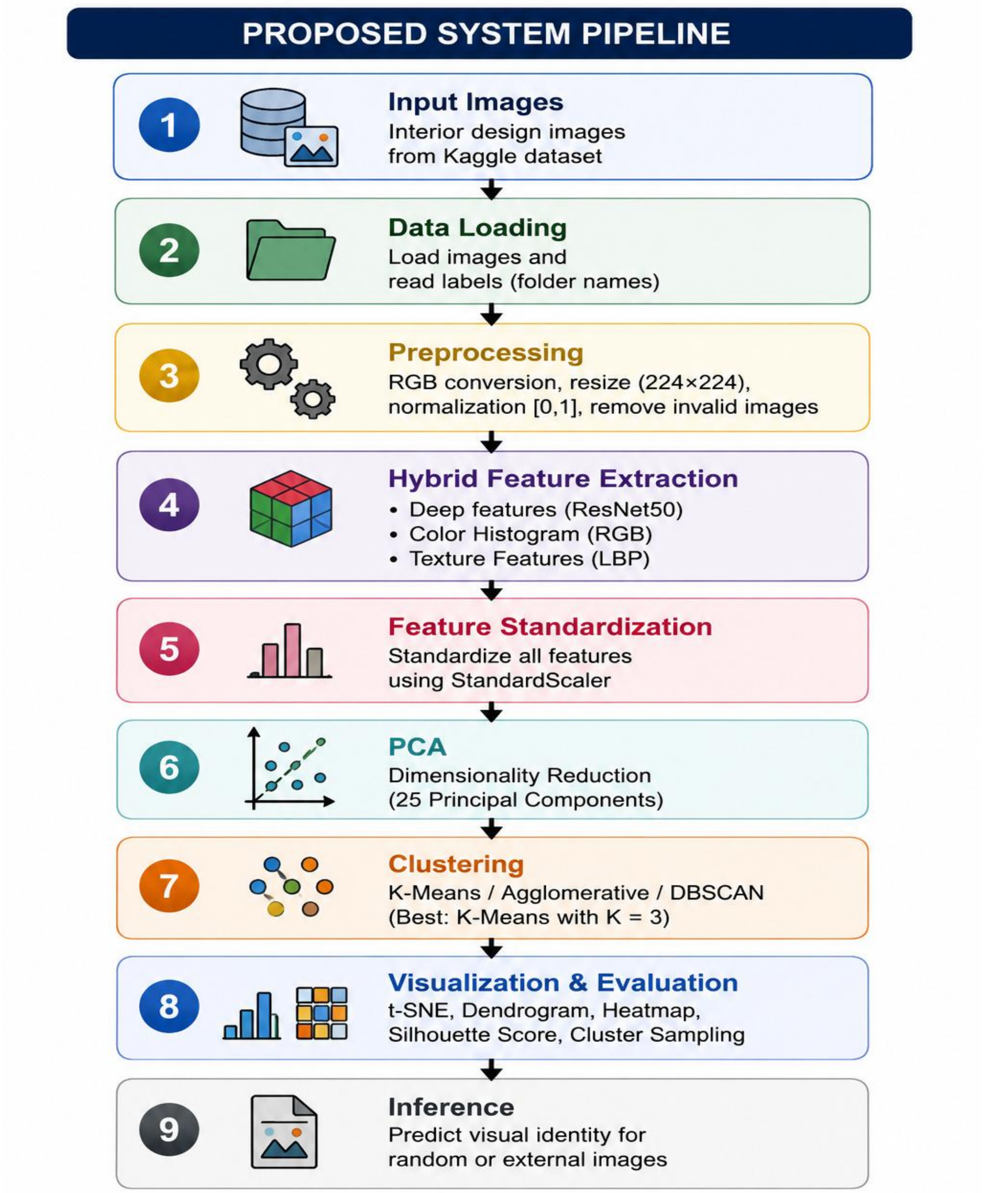


Figure 1: Overall pipeline of the proposed system

3.2 Dataset

The goal of this study was to identify visual patterns in static interior design images using a dataset collected from Kaggle. Pictures are grouped into various design styles in the dataset, which was downloaded from Kaggle. Each folder has a unique style.

The folder name of each picture serves as its label, which makes it easier to compare the results to the original categories later. A random sample of around 5,500 photographs was chosen after the data was shuffled in order to expedite experiments and shorten processing time.

Furthermore, the dataset balance was assessed by analyzing the distribution of images across various styles. The original image sizes were also examined, and the analysis showed that the images were consistent in size. However, preprocessing was still required to resize them into a fixed format suitable for feature extraction and deep learning models.

3.3 Preprocessing

The images were processed in several preprocessing stages before features were extracted.

To ensure that the colors were correct, the images were first loaded using OpenCV and then converted from BGR to RGB format. After that, every image was resized to 224 x 224 pixels, which is suitable for deep learning models.

The pixel values were then normalized to the range [0, 1] by dividing by 255. This enhances the stability and performance of the model.

During this procedure, any images that were either unreadable or invalid were deleted. The processed photos were ultimately saved as a NumPy file, facilitating speedier reuse without having to redo preprocessing.

3.4 Feature Extraction

To represent each image effectively, a hybrid feature extraction method was used. This method combines different types of features to capture more information from the images.

- Deep Features

The ResNet50 model was used to extract deep semantic features, capturing high-level visual information such as spatial structure, layout, and overall composition.

To increase speed and minimize memory usage, pictures were processed in batches. Each image was represented by a deep feature vector.

- Color Features

A 3D RGB histogram was used to extract color features. The distribution of colors in each picture is represented by this approach.

To ensure that the values in the histogram were consistent across all images, they were normalized. These characteristics aid in differentiating between styles that employ colors differently.

- Texture Features

With the Local Binary Pattern (LBP) approach, texture features were extracted.

To capture subtle texture patterns, LBP was applied to each image after it had been converted to grayscale. The data were then transformed into a histogram showing texture distribution.

This aids in distinguishing between various surface types, such as smooth, rough, or intricate textures.

- Hybrid Features

Finally, all features were combined into one feature vector, including:

- Deep features
- Color features
- Texture features

Color and texture features were scaled by 0.5 to balance their effect with deep features.

After that, all features were standardized to ensure they are on the same scale, which improves the performance of later steps such as clustering.

3.5 Principal Component Analysis (PCA)

After combining deep features, color histograms, and texture descriptors into one hybrid feature vector, the dimensionality of the data became very large. High-dimensional feature spaces may increase computational complexity and reduce clustering performance due to redundant or less informative variables. Therefore, Principal Component Analysis (PCA) was applied before clustering as a dimensionality reduction technique.

PCA transforms the original feature space into a smaller set of principal components while preserving the most important variance within the data. This helps simplify the representation space, reduce noise, and improve clustering efficiency.

Before applying PCA, all feature vectors were standardized using StandardScaler to ensure that each feature contributed equally to the transformation process. The dimensionality was then reduced to 25 principal components, which preserved the essential visual information while significantly simplifying the feature space.

Applying PCA helped simplify the feature space, reduce feature redundancy, and improve clustering efficiency. The reduced representation was later used as the input for all clustering algorithms.

3.6 Clustering

After dimensionality reduction, multiple unsupervised clustering algorithms were applied to group images according to visual similarity. The main objective of clustering was to discover hidden visual identities without relying on predefined labels.

Three clustering methods were explored and compared during experimentation:

- K-Means Clustering
- Agglomerative Hierarchical Clustering
- DBSCAN

K-Means Clustering

K-Means was selected as the primary clustering approach because of its simplicity, scalability, and effectiveness on compact visual representations.

The algorithm partitions the dataset into K clusters by minimizing the distance between feature vectors and their cluster centroids. Different K values were tested experimentally using the Silhouette Score to determine the most stable and interpretable clustering structure.

The best performance was achieved with K = 3, where the system successfully grouped images into three main visual identities. These clusters showed meaningful visual consistency in terms of layout, lighting, textures, and overall aesthetic composition.

Agglomerative Hierarchical Clustering

Agglomerative Clustering was also applied to analyze hierarchical relationships between images. This method initially treats each image as an individual cluster, then gradually merges the closest clusters based on feature similarity.

A dendrogram visualization was generated to better understand the structural relationships between the discovered visual identities. This helped provide additional interpretability and comparison with K-Means clustering results.

- ### DBSCAN

DBSCAN was explored as a density-based clustering algorithm capable of identifying irregular cluster structures and detecting outlier images.

Unlike K-Means, DBSCAN does not require defining the number of clusters in advance. In this project, DBSCAN was mainly used for outlier detection and identifying images that did not strongly belong to a specific visual identity.

However, the algorithm showed sensitivity to parameter selection due to the complexity and high dimensionality of the visual feature space.

4. Experimental Results and Discussion

4.1 Discussion

The preliminary unsupervised clustering phase provided useful insights into the underlying structure of the interior design dataset. By combining deep learning features with color and texture descriptors, the system grouped the images into three interpretable visual identities. Identity 0 showed noticeable alignment with Eclectic and Modern styles, while Identity 1 and Identity 2 were more closely associated with Transitional and Tropical aesthetics, respectively. In addition, the t-SNE visualization suggested a meaningful degree of separation between the discovered clusters, indicating that the hybrid feature representation was effective in capturing important stylistic patterns within the dataset.

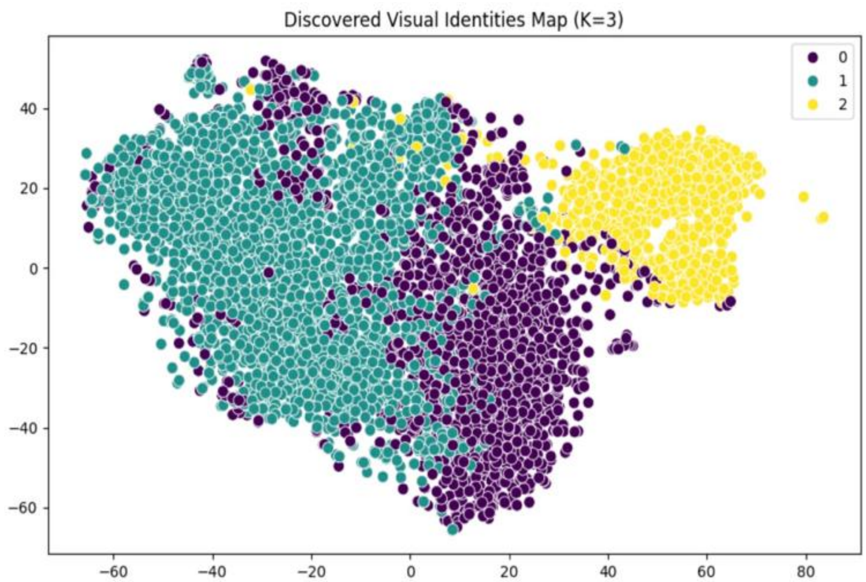
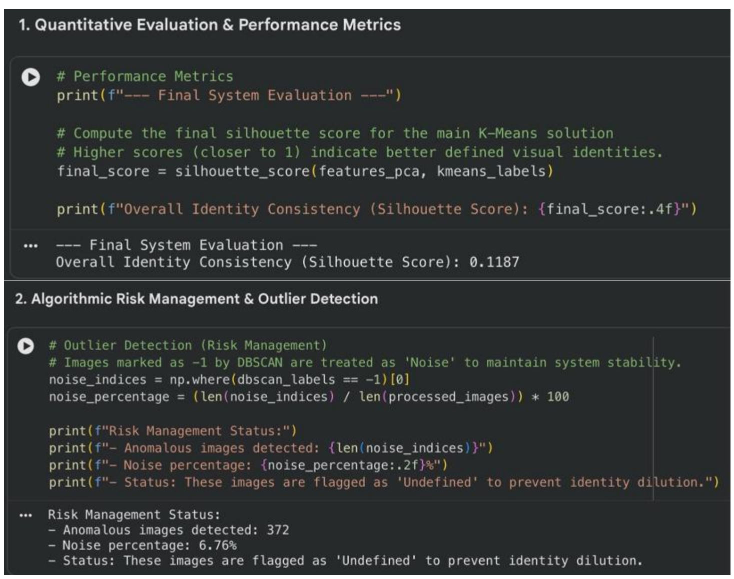


Figure 2: Discovered Visual Identities Map (K=3) visualizing the clustering distribution via t-SNE

4.2 Error Analysis

To ensure system reliability and investigate the quality of the discovered visual identities, a comprehensive error analysis was conducted. This evaluation relied on the Silhouette Score as a primary quantitative metric, where the system achieved a score of 0.1187. While this indicates measurable cluster formation, it also suggests modest separation due to the inherent visual overlap between diverse interior design styles.

To further investigate this behavior, DBSCAN was utilized as a diagnostic tool for algorithmic risk management and outlier detection. This process successfully identified 372 anomalous images (6.76% of the total samples) that did not strongly align with a specific visual identity. By flagging these samples as 'Undefined,' the system maintains its integrity and prevents identity dilution. These findings highlight that while the clusters are meaningful, further refinement of the feature space could enhance future separation and overall clustering stability.



1. Quantitative Evaluation & Performance Metrics

```python
# Performance Metrics
print(f"--- Final System Evaluation ---")

# Compute the final silhouette score for the main K-Means solution
# Higher scores (closer to 1) indicate better defined visual identities.
final_score = silhouette_score(features_pca, kmeans_labels)

print(f"Overall Identity Consistency (Silhouette Score): {final_score:.4f}")
```

```
--- Final System Evaluation ---
Overall Identity Consistency (Silhouette Score): 0.1187
```

2. Algorithmic Risk Management & Outlier Detection

```python
# Outlier Detection (Risk Management)
# Images marked as -1 by DBSCAN are treated as 'Noise' to maintain system stability.
noise_indices = np.where(dbscan_labels == -1)[0]
noise_percentage = (len(noise_indices) / len(processed_images)) * 100

print(f"Risk Management Status:")
print(f"- Anomalous images detected: {len(noise_indices)}")
print(f"- Noise percentage: {noise_percentage:.2f}%")
print(f"- Status: These images are flagged as 'Undefined' to prevent identity dilution.")
```

```
Risk Management Status:
- Anomalous images detected: 372
- Noise percentage: 6.76%
- Status: These images are flagged as 'Undefined' to prevent identity dilution.
```

Figure 3: Quantitative system evaluation results, including Silhouette Score and DBSCAN outlier detection metrics

4.3 Final Experimental Evaluation

To further evaluate the interpretability and visual consistency of the discovered clusters, additional qualitative experiments were conducted using representative image sampling, heatmap analysis, and hierarchical visualization techniques.

Representative image samples from each cluster revealed that images grouped within the same visual identity shared several common visual characteristics, including lighting style, spatial arrangement, texture composition, and color harmony. These observations further support the effectiveness of the proposed hybrid feature representation in capturing meaningful stylistic similarities between interior design images.

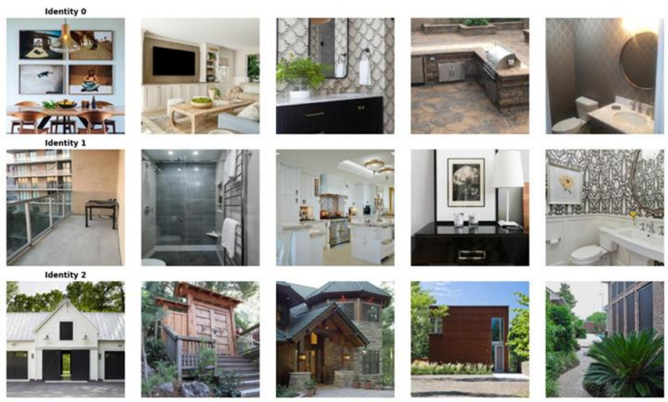


Figure 4: Representative image samples from the discovered visual identity clusters showing visual consistency within each identity

A heatmap analysis was also performed to compare the discovered visual identities with the original dataset style labels. The results demonstrated noticeable relationships between several discovered clusters and existing interior design categories. Some clusters showed stronger alignment with styles such as Modern, Eclectic, Transitional, and Tropical, supporting the interpretability of the clustering framework and its ability to capture broader visual patterns.

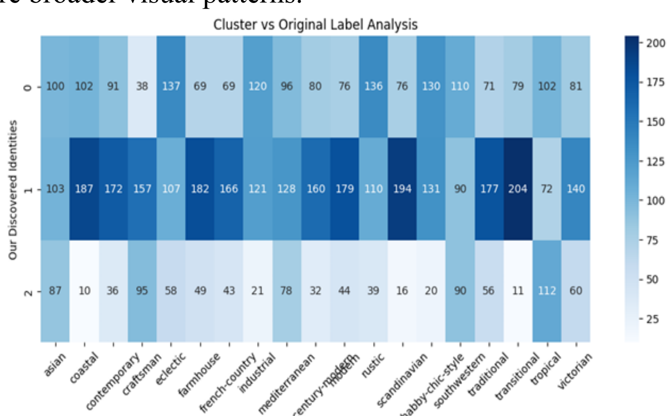


Figure 5: Heatmap comparison between discovered visual identities and original dataset style labels

To further analyze the structural relationships between the discovered visual identities, Agglomerative Hierarchical Clustering was used to generate a dendrogram visualization. The dendrogram illustrates how visually similar images gradually merge into larger hierarchical groups based on Euclidean distance, providing additional insight into the organization of the discovered image clusters.

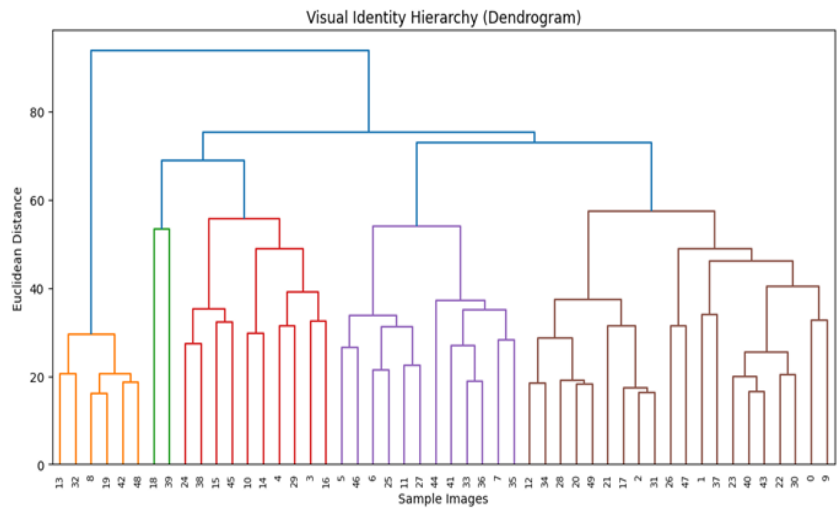


Figure 6: Dendrogram visualization generated using Agglomerative Hierarchical Clustering showing hierarchical relationships between image groups

Overall, the experimental results demonstrated that the proposed framework was capable of discovering visually meaningful and interpretable identities despite the complexity and overlap between interior design styles. The combination of quantitative evaluation and qualitative visualization techniques provided a more comprehensive understanding of the clustering behavior and the discovered visual patterns.

5. Comparison with Baselines

To evaluate the effectiveness of the proposed hybrid feature representation, multiple clustering algorithms were compared under the same experimental setup. All methods were applied to the identical PCA-reduced feature space (25 components) derived from the combined deep, color, and texture feature vectors. Three clustering algorithms were systematically evaluated: K-Means, Agglomerative Hierarchical Clustering, and DBSCAN. Each algorithm was assessed based on quantitative metrics (Silhouette Score), cluster balance (distribution of samples across clusters), and qualitative visual coherence of the resulting groups.

Clustering Method	Silhouette Score	Cluster Distribution	Key Characteristics
K-Means (K = 3)	0.1187	Balanced across 3 clusters	Best overall separation; visually consistent identities; centroid-based optimization produced well-defined stylistic groupings.
Agglomerative Hierarchical Clustering (K = 3)	0.1124	Less stable in some trials	Captured hierarchical relationships; useful for dendrogram visualization and structural analysis; less stable than K-Means.
DBSCAN (eps = 3, min_samples = 5)	0.1098	Uneven with 372 noise points (6.76%)	Effective for outlier detection and risk management; identified ambiguous images as noise; less suitable for identity discovery due to sensitivity to parameter selection.

K-Means produced well-balanced visual identities with the highest Silhouette Score and the clearest separation between clusters, making it the primary clustering method adopted in this project. Its centroid-based optimization enabled effective grouping of images with similar semantic and stylistic characteristics. Agglomerative Clustering provided an alternative perspective through hierarchical structure analysis and dendrogram visualization, which supported the interpretation of relationships between the discovered identities. DBSCAN was mainly utilized for outlier detection and risk management. The algorithm successfully identified anomalous or undefined images as noise, flagging 372 samples that did not strongly belong to any single visual identity, thereby supporting the robustness of the overall clustering pipeline.

These results indicate that K-Means, when combined with the proposed hybrid feature representation, is the most suitable clustering approach for discovering emergent visual identities in interior design images.

6. Identified Limitations

Despite the promising results, several limitations were encountered during the experimental phase, which provide important context for interpreting the findings:

- **Stylistic Overlap and Metric Sensitivity:** One of the main challenges was the relatively modest Silhouette Score. This is primarily due to the visual overlap between many interior design styles, where different categories share similar colors, layouts, and textures, making numerical separation difficult.
- **Algorithmic Consistency:** Inconsistency in behavior across different algorithms was observed. While K-Means produced balanced clusters, Agglomerative Clustering was often unstable, and DBSCAN proved to be more effective for outlier detection than for primary clustering.
- **Quantitative vs. Qualitative Gap:** Quantitative metrics alone were insufficient to fully reflect cluster quality. In several cases, the visual groupings appeared meaningful despite low numerical scores, necessitating a greater reliance on qualitative analysis, such as t-SNE visualization and heatmap interpretation.
- **Computational Constraints:** The feature extraction process—combining deep features, color histograms, and texture descriptors—was computationally expensive. This required using a reduced subset of the dataset to maintain practical experimentation and system stability.

7. Conclusion and Future Work

This project presented an unsupervised computer vision system for discovering emergent visual identities in static interior design images. By combining deep features extracted from ResNet50 with handcrafted color histogram and LBP texture descriptors, the system was able to build a rich hybrid representation of each image. After applying PCA and multiple clustering techniques, K-Means produced the most balanced and interpretable clustering result.

Although the silhouette score was relatively modest, the qualitative results showed that the discovered clusters still captured meaningful visual patterns. The project demonstrated that unsupervised learning can be used to reveal broader aesthetic structures and latent visual relationships that are not always reflected by predefined style labels. Overall, the system provided a practical and interpretable approach for organizing visually complex image collections.

For future work, this project can be extended from a visual clustering system into a more intelligent framework for interpreting the discovered patterns. More advanced interpretability mechanisms could be incorporated to explain why certain images belong to specific clusters, allowing the system not only to assign a visual identity, but also to clarify the factors influencing that assignment.

Another possible direction is to connect the system to a practical interior design application. Instead of only clustering images, the model could be used to suggest similar design styles or help users explore visual identities that best match their preferences. This would transform the project from an experimental model into a more practical and user-oriented tool.

A further extension would be to broaden the scope of analysis to include other types of design-related images, such as furniture images, architectural spaces, or interior facades. This would help evaluate whether the system can discover emergent visual identities across a wider range of visual design domains.

8. Task Distribution

Team Member	Main Contribution
Danah	Contributed to the early conceptual stages of the project, including problem definition, dataset description, and parts of the introduction and background writing. Also worked on visualization and qualitative analysis, including the t-SNE visual map, cluster-label heatmap, representative identity sampling, and support for the conclusion, future work, and contribution statement.
Sadeem	Contributed to the planning stage through algorithm selection, task distribution, and expected challenges. During implementation, focused on clustering models, including K-Means optimization, Agglomerative Clustering, dendrogram generation, and comparison with baseline methods.
Meran	Contributed to the high-level system design and implementation planning. During development, worked on feature representation analysis, evaluation, testing, and inference demos, including validation on unseen images and interpretation of performance results.
Retal	Contributed to the methodology design and expected project contribution during the planning stage. In implementation, focused on hybrid feature extraction by combining deep learning features with handcrafted descriptors, as well as dimensionality reduction and clustering-related experimentation.
Mariam	Contributed to project motivation, evaluation planning, and risk analysis in the early stages. During implementation, worked on dataset preparation and preprocessing, including image cleaning, resizing, normalization, output organization, and support for documentation and results discussion.

9. References

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